

A sharp additive-three cover bound for intersecting uniform hypergraphs

Provisional mathematical authors: Rachel Teo, Wenyi Wang, Hamdi Abderrahmene

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<https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr> ;

<https://www.sundai.club/events/boston/recursive-learning-hack-with-harvard-innovation-labs>

Rachel Teo - Sakana AI.

Wenyi Wang - Sakana AI.

Hamdi Abderrahmene - Sakana AI.

Affiliations follow the recorded author declarations or public institutional/profile sources. Preferred author names, author order and publication affiliations await author review. This editorial compilation does not establish anthology coauthorship.

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Central source and adjudication archive: <https://github.com/alejandrozu/openmath-2026-judging>

Abstract

For every finite intersecting r -uniform hypergraph H , $4\tau(H) \leq |H| + r + 3$. The degree-at-most-three residual bound closes the explicit additive-three question in Sivashankar's pre-event paper; degree-four peeling yields the unrestricted form and $|H| \geq 3r - 3$ whenever $\tau(H) = r$.

Statement and relation to the existing question

For a finite hypergraph H , $\tau(H)$ is the minimum size of a vertex set meeting every edge. Intersecting means that every pair of edges has a common vertex. The new residual theorem asserts $4\tau(J) \leq |J| + r + 3$ for an intersecting r -uniform J in which every vertex lies in at most three edges.

The source paper proves the analogous additive-four estimate and explicitly asks whether three is possible. The submitted proof answers that subsidiary question. A standard peeling argument then proves $4\tau(H) \leq |H| + r + 3$ without the degree restriction. When $\tau(H) = r$, rearrangement gives $|H| \geq 3r - 3$, and in particular $|H| \geq 21$ at $r = 8$. The triangle example supplies sharpness of the additive constant.

The residual packing improvement

The residual proof begins with the source paper's matching and witness machinery. A maximum matching organizes the edges and the uncovered remainder. Its associated link graphs form a three-link packing, encoding the ways covered vertices can meet the uncovered ground set.

The older count leaves a potential equality case consistent with additive four. The new coloured-link budget proves a strict support-sum inequality when the uncovered ground set is odd and large enough relative to the number of packing blocks. The proof separates small configurations and develops structural/equality exclusions for the remaining coloured packings. Combining that strict

inequality with the original selected-count identity rules out exactly the obstruction to additive three.

SharpEndpoint initially presents the packing inequality as an explicit argument; the final Main endpoint supplies it from the Packing and Colored modules. This distinction matters: reading the bridge alone could suggest an assumed lemma, while the selected final closure is intended to be unconditional. The independent archived replay reports 18 Main-closure modules and a separate endpoint-definition audit, without importing the Kahn-dependent asymptotic section.

Peeling and the general consequence

If some vertex v has degree at least four, delete every edge containing v and call the remaining hypergraph H' . It stays intersecting and r -uniform. A minimum cover of H' , together with v , covers H , so $\tau(H) \leq \tau(H') + 1$. The deletion removes at least four edges, exactly compensating for the factor four in the cover inequality. Induction on the number of edges lifts the residual additive-three bound to all H .

If no vertex has degree at least four, the residual theorem applies directly. This yields the unrestricted cover inequality and its lower-bound corollaries in the same family. The $r=8$ specialization and triangle sharpness should be presented as consequences, not independent breakthroughs.

Attribution, scope and publication

Varun Sivashankar's pre-event paper supplies the $+4$ theorem, matching/witness infrastructure and sharpness examples. Its ancillary Lean prefix is reused with explicit attribution and documented compatibility adjustments. The proposed original delta is the strict packing improvement to $+3$ and the resulting all- r elementary bound.

The original $O(r)$ Erdős 21 question was already resolved by Kahn, and the source paper's stronger approximately $3.053r$ asymptotic lower coefficient is not improved here. A title claiming a full solution of Erdős 21 would be misleading. A focused paper answering the explicit residual question, with the supplementary Lean closure and clear source attribution, is the right publication form.

This short manuscript records the construction and selected endpoints. The companion Original results anthology contains the extended ordinary proof exposition and its explicitly stated premises; the preserved Lean source remains the complete formal proof record.

Verification and reproducibility

Fresh execution: sakana-erdos21-sharp-residual: PASS; 18/18 custom modules compiled successfully; receipt Verification/sakana-erdos21-sharp-residual-fresh-build.json. Compiler pins, source hashes and endpoint axiom outputs are in Verification. Historical receipts and finite checks do not replace fresh replay.

ErdosLovasz.residual_cover_bound_sharp

```
theorem residual_cover_bound_sharp (r : ℕ) (J : Finset (Finset V))
(huni : IsUniform r J) (hint : Intersecting J)
(hdeg : ∀ v, edgeDeg J v ≤ 3) :
4 * coverNum J ≤ J.card + r + 3
```

Source: proofs/erdos21-sharp-residual/OpHack/Erdos21/Main.lean:25.

Source SHA-256: 9314ae7ba95b6fd9922ccfea882f1d33a6c11c9a40d750109014c8184ab7e414

ErdosLovasz.cover_bound_sharp

```
theorem cover_bound_sharp (r : ℕ) (H : Finset (Finset V))
(huni : IsUniform r H) (hint : Intersecting H) :
4 * coverNum H ≤ H.card + r + 3
```

Source: proofs/erdos21-sharp-residual/OpHack/Erdos21/Main.lean:33.

Source SHA-256: 9314ae7ba95b6fd9922ccfea882f1d33a6c11c9a40d750109014c8184ab7e414

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